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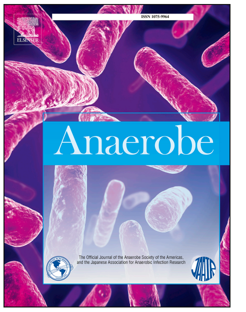

Toxinotyping and molecular characterization of antimicrobial resistance in *Clostridium perfringens* isolated from different sources of livestock and poultry

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PII: S1075-9964(20)30154-2

DOI: https://doi.org/10.1016/j.anaerobe.2020.102298

Reference: YANAE 102298

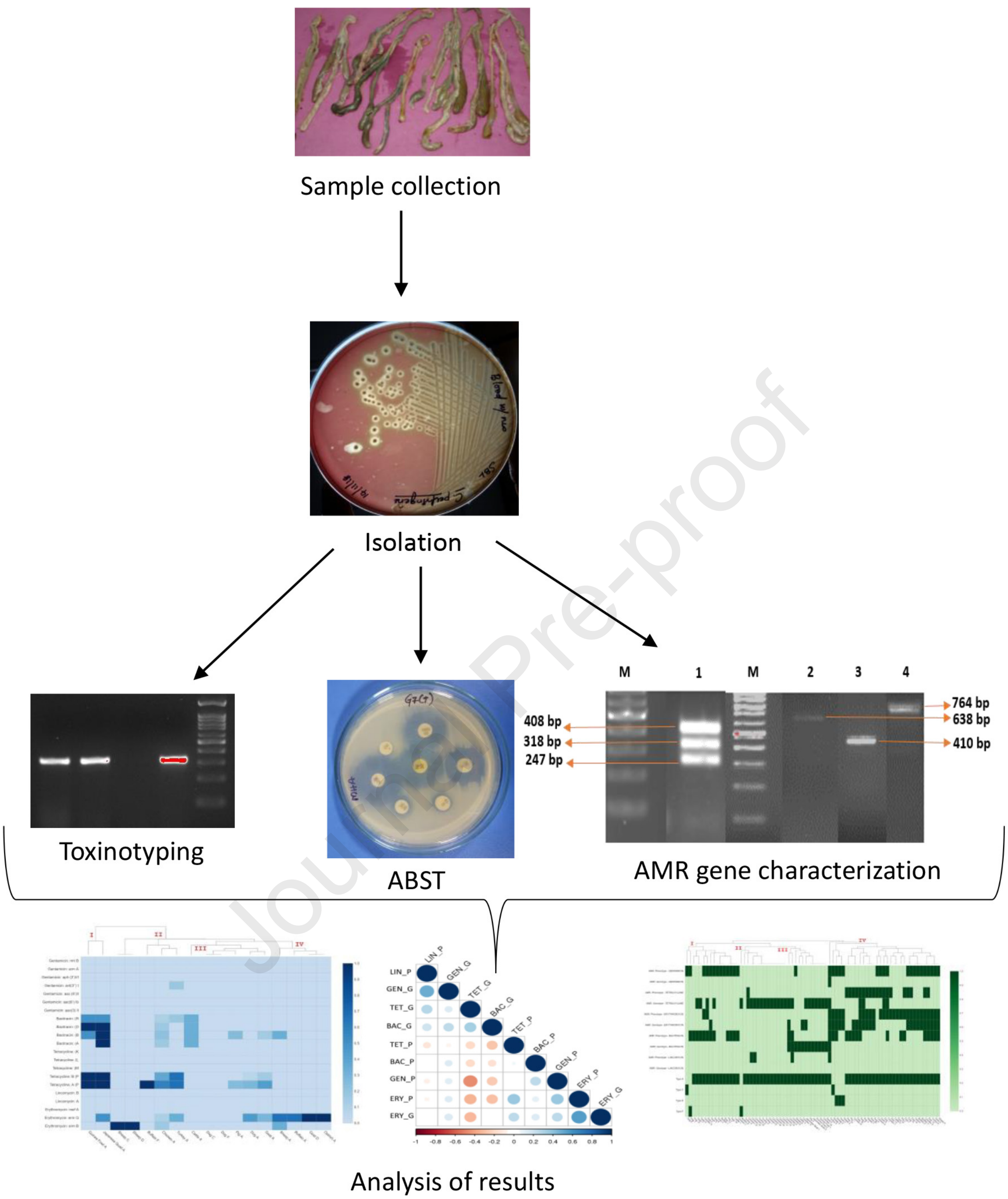
To appear in: *Anaerobe*

Received Date: 23 May 2020

Revised Date: 6 November 2020

Accepted Date: 15 November 2020

Please cite this article as: Anju K, Karthik K, Divya V, Mala Priyadharshini ML, Sharma RK, Manoharan S, Toxinotyping and molecular characterization of antimicrobial resistance in *Clostridium perfringens* isolated from different sources of livestock and poultry, *Anaerobe*, https://doi.org/10.1016/j.anaerobe.2020.102298.



Sample collection
Isolation
Toxinotyping
ABST
M
1
M
2
3
4
408 bp
318 bp
247 bp
764 bp
638 bp
410 bp
AMR gene characterization
LIN_P
GEN_G
TET_G
BAC_G
TET_P
BAC_P
GEN_P
ERY_P
ERY_G
Analysis of results

Toxinotyping and molecular characterization of antimicrobial resistance in *Clostridium perfringens* isolated from different sources of livestock and poultry

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ABSTRACT

The present study was designed to understand the presence of antimicrobial resistance among the prevalent toxinotypes of *Clostridium perfringens* recovered from different animals of Tamil Nadu, India. A total of 75 (10.76%) *C. perfringens* were isolated from 697 multi-species fecal and intestinal content samples. *C. perfringens* type A (90.67%), type C (2.67%), type D (4%) and type F (2.67%) were recovered. Maximum number of isolates were recovered from dog (n= 20, 24.10%) followed by chicken (n=19, 5.88%). Recovered isolates were resistant to gentamicin (44.00%), erythromycin (40.00%), bacitracin (40.00%), and tetracycline (26.67%), phenotypically and most of the isolates were found to be resistant to multiple antimicrobials. Genotypic characterization revealed that tetracycline (41.33 %), erythromycin (34.66%) and bacitracin (17.33%) resistant genes were present individually or in combination among the isolates. Combined results of phenotypic and genotypic characterization showed the highest percentage of erythromycin resistance (26.66%) among the isolates. None of the isolates showed amplification for lincomycin resistance genes. The correlation matrix analysis of genotypic resistance showed a weak positive relationship between the tetracycline and bacitracin resistance while a weak negative relationship between the tetracycline and erythromycin resistance. The present study thus reports the presence of multiple-resistance genes among *C. perfringens* isolates that may be involved in the dissemination of resistance to other bacteria present across species. Further insights into the genome can help to understand the mechanism involved in gene transfer so that measures can be taken to prevent the AMR spread.

Keywords: *Clostridium perfringens*, toxinotype, antimicrobial resistance, resistance gene, prevalence, India

1. Introduction

Clostridium perfringens is a ubiquitous organism found in the normal intestine, sewage, food products, etc. *C. perfringens*, a gram-positive, anaerobic spore-forming organism is also associated with various diseases in animals and human [1,2]. *C. perfringens* produces different extracellular enzymes and toxins. Based on the major toxins produced namely α-toxin, β-toxin, ε-toxin and ι-toxin, *C. perfringens* have been classified into 5 toxinotypes (A to E) [3]. Recently, two more toxinotypes (F and G) has been proposed based on additional toxin repertoire. It was proposed that type F include *C. perfringens* producing enterotoxin (*cpe* gene) involved in food poisoning and type G include *C. perfringens* producing NetB toxin (*netB* gene) involved in necrotic enteritis in poultry [4].

C. perfringens type A causes enterocolitis in horses, pigs and dogs. Apart from type G, type A is also involved in necrotic enteritis in chickens. Type B causes necrotizing enteritis in cattle, sheep. Type C causes necrohemorrhagic enteritis in sheep, goats, piglets, horses, etc. Type D causes enterotoxaemia in cattle, sheep, goat and type E causes enteritis in cattle, sheep and rabbits. Type F causes hemorrhagic gastroenteritis in dogs and colitis in horse. Type F is mainly responsible for human food poisoning, antibiotic associated diarrhea and sporadic diarrhea. Type G is involved in necrotic enteritis in poultry [5]. Enterotoxaemia caused by type D in sheep and goat is a more prevalent and important disease that can cause sudden mortality [5].

Antimicrobial agents like tetracycline, lincomycin, bacitracin and other agents are commonly used in feed to improve health and production of poultry and other animals [6,7]. Long term use or unintended use of antimicrobial agents has led to the rapid development of resistance among *C. perfringens* isolates worldwide [8]. Antimicrobial resistance is conferred by various genes and they play a major role in transfer of resistance to other bacteria within the genus or between genera Understanding the role of the resistant genes can aid in the prevention of gene transfer between the microbes. Genes like *tetP*(B), *tet*(M), *tetA*(P), and *tetB*(P) were linked with tetracycline resistance which has been reported from various parts of the world [9]. Similarly, other genes have been identified and are linked to antimicrobial resistance [10].

Antimicrobial resistance among *C. perfringens* isolates recovered from India has been documented by various researchers [11,12]. Most of the studies from India on antimicrobial resistance among *C. perfringens* were restricted to phenotypic testing. Reports on genotypic characterization of antimicrobial resistance genes among *C. perfringens* from India are scarce. A comprehensive study on the prevalence of *C. perfringens* from different animal species, its antimicrobial resistance determination both by phenotypic and genotypic testing can help to understand the role of the genes in resistance transfer. Therefore, the present study was designed to know the prevalence of *C. perfringens* among different animal species and its antimicrobial resistance both by phenotypic and genotypic methods.

2. Material and methods

2.1. Sample collection

A total of 697 samples were collected randomly from four different districts of Tamil Nadu (Table 1). Fecal swabs were collected from Kanchipuram (n=100), Vellore (n= 91), Chennai (n=83, dog samples), Tiruvallur (n=207) while intestinal contents were collected during post-mortem (PM) from Chennai (n=116, chicken samples) and Tiruvallur (n=100, chicken samples). Sterile swabs were used for the collection of feces sample and intestinal contents were collected in sterile containers during PM examination of animals and birds. Animals subjected to PM examination at the Central University Laboratory, TANUVAS with various ailments were used for the collection of intestinal contents. All sheep and goat from the Vellore district had a history of diarrhea. Similarly, all dogs included in the study from the Chennai district had gastrointestinal disorders. Data on the health status of other animals in the study is unknown. The samples were transported immediately to the laboratory within 24 hours of collection.

Table 1: Samples collected from different species of animals and birds of various sources

S. No.	Place (District)	Species (Numbers)	Total

			(Numbers)
1	Kanchipuram	Goat (17), Sheep (30), Cattle (19), Buffalo (7), Ostrich (2), Pig (7), Chicken (15), Rabbit (3)	100
2	Tiruvallur	Chicken (192), Turkey (7), Guinea fowl (5), Buffalo (9), Pig (16), Japanese Quail (6), Goat (33), Sheep (21), Cattle (18)	307
3	Vellore	Goat (67), Sheep (24)	91
4	Chennai	Dog (83), Poultry (116)	199
Total			**697**

2.2. Isolation and identification of C. perfringens

Robertson's cooked meat (RCM) medium (HiMedia, India) was used for the enrichment of the samples to isolate *C. perfringens.* Samples were inoculated in the RCM medium and heated to 70 ºC for 15 min. Later, the broth was incubated at 37°C for 24 - 48 hours under anaerobic condition in McLintock anaerobic jar using a gas pack (Anerogas pack, HiMedia, India). After 24- 48 hours, the turbid broth was cultured on 5% sheep blood agar with neomycin (200 µg/ml) and Tryptose Sulphite Cycloserine (TSC) agar (HiMedia, India) to isolate a pure culture of *C. perfringens*. Colonies showing double zone of hemolysis (indicative of *C. perfringens*) on blood agar and black colonies on TSC agar were Gram's stained for microscopic examination. Colonies showing Gram-positive bacilli were used for further molecular characterization. Pure, confirmed *C. perfringens* isolates were preserved in 20% glycerol in brain heart infusion broth and stored in -80°C.

2.3. Toxinotyping of the C. perfringens isolates

All the recovered isolates were used for molecular characterization employing established primers. DNA of the recovered isolates was extracted by the snap chill method. Briefly, 3-5 colonies were suspended in 100 µl of nuclease-free water, boiled for 10 min and frozen for 5 min then centrifuged at 12,000 rpm for 5 min. The supernatant was used as DNA for further molecular characterization. DNA of *C. perfringens* type D field isolate already available in the laboratory was used as a positive control during all the PCR reactions. Initially, to confirm isolates as *C. perfringens,* α-toxin (*cpa*) gene primers were used. Alpha toxin positive *C. perfringens* DNA was toxinotyped by multiplex PCR targeting genes viz. β-toxin (*cpb*), β2-toxin (*cpb2*), ε-toxin (*etx*), ι-toxin (*iap*), enterotoxin (*cpe*). All the primers and protocol for PCR was followed as per van Asten et al. [13] (Supplementary file 1). PCR targeting the *netB* gene was carried out as per Bailey et al. [14]. Amplified products were subjected to agarose (1.5%) gel electrophoresis and visualized under Gel Doc^{TM} XR (Bio-Rad, USA).

2.4. Antimicrobial susceptibility testing

Antimicrobial susceptibility testing (AST) was carried out using 8 different agents including Tetracycline (TE, 30 µg), Gentamicin (GEN, 10 µg), Erythromycin (E, 15 µg), Vancomycin (VA, 10 µg), Rifampicin (RIF, 5µg), Penicillin (P, 2 units), Bacitracin (B, 10 units), Lincomycin (L, 2 µg) and Enrofloxacin (EX, 5 µg). These antimicrobials were selected based on their use in animals and also based on the previous literature regarding its resistance among *C. perfringens* isolates. All the AST discs were procured from HiMedia, India. The disk diffusion method was used for AST of all the 75 isolates as per the guidelines provided by Clinical Laboratory Standards Institute [15] and *Staphylococcus aureus* was used as the reference since there are no breakpoints for *C. perfringens*.

2.5. Antimicrobial resistance (AMR) gene detection

AST results were used as base data to select AMR genes for PCR. Since tetracycline, gentamicin, erythromycin and bacitracin were found to be resistant in many isolates and lincomycin was least resistant in the present study, these antimicrobials were selected for AMR gene identification by PCR. PCR was carried out employing published primers targeting 21 genes for the selected antimicrobial agents. DNA was extracted from all the 75 positive isolates using DNeasy blood and tissue kit (Qiagen, Germany) as per the manufacturer's instruction. Genes targeted for PCR were erythromycin-resistant genes [*erm*(B), *erm*(Q), *mef*(A)], bacitracin-resistant genes [*bcrA*, *bcrB*, *bcrD*, *bcrR*], tetracycline-resistant genes [*tetA*(P), *tetB*(P), *tet*(M), *tet*(L), *tet*(K)], lincomycin- resistant genes [*lnu*(A), *lnu*(B)] and gentamicin-resistant genes [*aac*(3)-ll, *aac*(6')-lb, *aac*(6')-ll, *ant*(3'')-l, *aph* (3')-VI, *arm* A, *rmt* B]. The primers used for the detection of these resistance genes are provided in the supplementary file 2. All the primers and protocols for PCR were followed as per earlier published protocols (Table S2). Amplified products were subjected to agarose (1.5%) gel electrophoresis and visualized under Gel Doc™ XR (Bio-Rad, USA).

2.6. Data analysis

Antimicrobial resistance profile (phenotypic and genotypic) was converted into binary coding (positive/resistant = 1, negative/susceptible = 0). Hierarchical clustering was designed and analyzed using Displayr software (https://app.displayr.com/). The correlation matrix between phenotypic and genotypic antimicrobial resistance was created and analyzed using Statistical tools for high-throughput data analysis online software (www.sthda.com).

3. Results

3.1. Isolation and toxinotyping

Out of 697 samples collected, 75 samples (10.76%) were positive for *C. perfringens cpa* gene by PCR. Among the 75 isolates, 67 (89.33%) isolates belong to Type A (*cpa* gene positive) and one

(1.33%) isolate was positive for *cpa* and *cpb2* gene. Two isolates (2.67%) belonged to type C (*cpa* and *cpb* gene), 3 (4.0%) isolates belonged to type D (*cpa* and *etx* gene) and 2 (2.67%) isolates belonged to type F (*cpa* and *cpe* gene) (Figure 1). *iap* and *netB* genes were found negative in all the isolates tested.

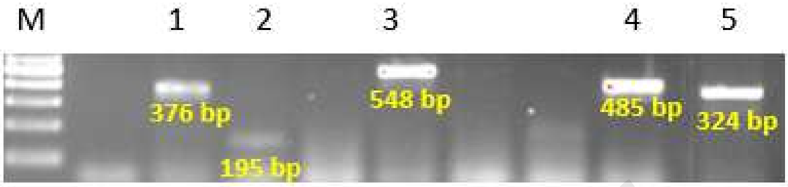

Fig.1 Agarose gel picture showing PCR detection of *C. perfringens* toxins.
M: 100 bp ladder, **Lane 1:** ε toxin (376 bp); **Lane 2:** β toxin (195 bp), **Lane 3:** enterotoxin (548 bp); **Lane 4:** β2 toxin (485 bp); **Lane 5:** α toxin (324 bp).

Toxinotyping revealed that all poultry (chicken, turkey, guinea fowl, ostrich and Japanese quail) isolates (n=30) were type A. Among the 30 isolates, one isolate from chicken showed positive for *cpa* and *cpb2* gene. Similarly, a total of 38 isolates from the dog (n= 18), goat (n= 7), sheep (n=3), pig (n=4), cattle (n= 3) and buffalo (n=3) belonged to type A. One isolate from each dog and sheep were type C, one isolate from goat and two isolates from sheep belonged to type D. One isolate each from dog and buffalo were type F (Table 2).

Table 2. Species wise classification of toxinotypes identified in the study

Species	**Total**	***C. perfringens* Positive (Nos.)**	***C. perfringens* (%)**	**Toxinotypes, numbers (%)**				
				Type A		**Type C**	**Type D**	**Type F**
				cpa	***cpa+cpb2***	***cpa+cpb***	***cpa+etx***	***cpa+cpe***
Chicken	323	19	5.88	18(94.74)	1 (5.26)	0	0	0
Ostrich	2	1	50.00	1 (100)	0	0	0	0
Turkey	7	6	85.71	6 (100)	0	0	0	0
Guinea Fowl	5	2	40.00	2 (100)	0	0	0	0
Japanese Quail	6	2	33.33	2 (100)	0	0	0	0
Dog	83	20	24.10	18 (90)	0	1 (5.0)	0	1 (5.0)
Goat	120	8	6.67	7 (87.5)	0	0	1 (12.5)	0
Sheep	75	6	8.00	3 (50)	0	1 (16.67)	2 (33.33)	0
Pig	23	4	17.39	4 (100)	0	0	0	0
Cattle	37	3	8.11	3 (100)	0	0	0	0
Buffalo	16	4	25.00	3 (75)	0	0	0	1 (25)
Total	**697**	**75**	**10.76**	**67 (89.33)**	**1 (1.33)**	**2 (2.67)**	**3 (4.0)**	**2 (2.67)**

Total of 9 (9%) *C. perfringens* isolates were recovered from Kanchipuram, 35 (11.40%) from Tiruvallur, 9 (9.89%) from Vellore and 22 (11.05%) from Chennai (table 3). All isolates from the Kanchipuram district belonged to type A. One sheep isolate from Vellore and one dog isolate from Chennai belonged to type C while two sheep isolates from Vellore and one goat isolate from Tiruvallur belonged to type D. One buffalo isolate from Tiruvallur and one dog isolate from Chennai belonged to type F. One chicken isolate from Tiruvallur was found positive for *cpb2* gene. All the other isolates from Tiruvallur, Vellore and Chennai belonged to type A (Table 3).

Table 3. District wise classification of positive *C. perfringens* isolates

S. No.	District	Species											Total
		Cattle	Buffalo	Sheep	Goat	Pig	Chicken	Ostrich	Japanese Quail	Turkey	Guinea fowl	Dog	
1	Kanchipuram	2	3	0	0	3	0	1	0	0	0	0	**9**
2	Tiruvallur	1	1	1	8	1	12	0	2	6	2	1	**35**
3	Vellore	0	0	5	4	0	0	0	0	0	0	0	**9**
4	Chennai	0	0	0	0	0	3	0	0	0	0	19	**22**
	Total	**3**	**4**	**6**	**12**	**4**	**15**	**1**	**2**	**6**	**2**	**20**	**75**

3.2. Antimicrobial susceptibility testing

Among the 75 isolates, phenotypically 26.66% of isolates showed resistance against tetracycline, 40% of the isolates were resistant to erythromycin and bacitracin. A higher percentage of isolates (44%) showed gentamicin resistance phenotypically. *C. perfringens* isolates showed lower resistance to vancomycin and lincomycin antibiotics, respectively. Several isolates showed multi- drug resistance and the results are presented in Table 4.

Table 4. Phenotypic antimicrobial resistance pattern of the isolates obtained in this study

Species	***C. perfringens* Positive (Nos.)**	**Gentamicin (%)**	**Tetracycline (%)**	**Erythromycin (%)**	**Rifampicin (%)**	**Penicillin (%)**	**Enrofloxacin (%)**	**Bacitracin (%)**	**Vancomycin (%)**	**Lincomycin (%)**
Chicken	19	5 (26.31)	5 (26.31)	3 (15.79)	5 (26.32)	3 (15.79)	4 (21.05)	7 (36.84)	2 (10.53)	3 (15.79)
Ostrich	1	1 (100)	0	1 (100)	0	0	0	0	0	0
Turkey	6	2 (33.33)	0	2 (33.33)	0	1 (16.67)	0	2 (33.33)	0	1 (16.67)
Guinea Fowl	2	0	0	0	0	0	0	0	0	0
Japanese Quail	2	0	0	0	1 (50.00)	0	0	0	0	0
Dog	20	10 (50.00)	12 (60.00)	12 (60.00)	5 (25.00)	1 (5.00)	1 (5.00)	10 (50.00)	1 (5.00)	0
Goat	8	5 (62.50)	1 (12.50)	3 (37.50)	0	1 (12.50)	1 (12.50)	4 (50.00)	0	0
Sheep	6	5 (83.33)	1 (16.67)	5 (83.33)	0	3 (50.00)	0	3 (50.00)	0	0
Pig	4	1 (25.00)	1 (25.00)	1 (25.00)	2 (50.00)	0	1 (25.00)	3 (75.00)	0	0
Cattle	3	1 (33.33)	0	1 (33.33)	0	3 (100.00)	2 (66.67)	1 (33.33)	2 (66.67)	0
Buffalo	4	3 (75.00)	0	2 (50.00)	0	0	0	0	0	0
Total	**75**	**33 (44.00)**	**20 (26.67)**	**30 (40.00)**	**13 (17.33)**	**12 (16.00)**	**9 (12.00)**	**30 (40.00)**	**5 (6.67)**	**4 (5.33)**

3.3. Antimicrobial resistance (AMR) gene pattern

Tetracycline resistance genes *tetA*(P) and *tetB*(P), either in combination or alone was detected in 31 isolates of *C. perfringens* (41.33%) of which only 5 isolates (6.66%) showed tetracycline resistance phenotypically. None of the isolates were found positive for *tet*(M), *tet*(L) and *tet*(K) genes. Only one isolate harbored gentamicin resistance gene (*ant*(3'')-l) which was found to express phenotypic resistance. Among bacitracin genes, 3 isolates were found to contain all the four bacitracin resistance genes (*bcrA*, *bcrB*, *bcrD and bcrR*) and 10 isolates detected with either one or combination of these genes. Three isolates positive for the bacitracin gene genotypically were also found resistant phenotypically. A total of 20 out of 26 isolates detected with erythromycin resistance genes (*erm*(Q) and *erm*(B)) expressed resistance phenotypically. A higher number of isolates (26.66%) showed erythromycin resistance both genotypically and phenotypically. None of the isolates had amplification for lincomycin resistance genes. The number of *C. perfringens* isolates detected with each resistance genes are depicted in Table 5 and Figure 2.

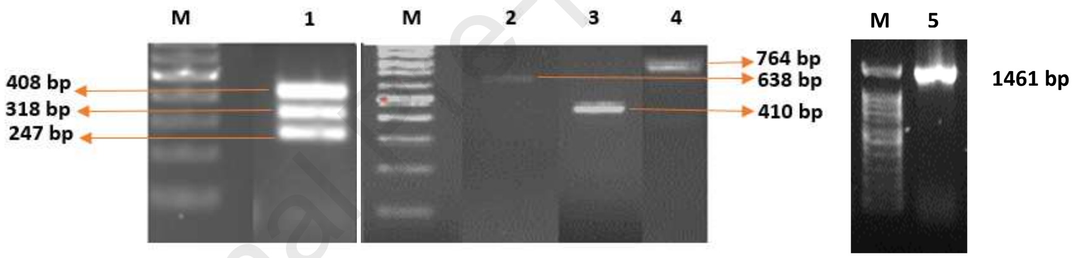

Fig.2. Agarose gel picture showing PCR detection of antimicrobial resistant genes. M: 100 bp ladder, **Lane 1:** Bacitracin resistant genes *bcrA* [408 bp], *bcrB* [247 bp], *bcrD* [318 bp]; **Lane 2:** Erythromycin resistant gene *erm*(B) [638 bp]; **Lane 3:** Erythromycin resistant gene *erm*(Q) [410 bp]; **Lane 4:** Tetracycline resistant gene *tetA* (P) [764 bp]. Lane 5: *tetB*(P) [1461 bp]

Table 5. PCR results of antimicrobial resistance genes

S. No.	Antimicrobial resistance gene	Number of isolates (%)
Erythromycin		
1	*erm*(B)	5 (6.67)
2	*erm*(Q)	19 (25.33)
3	*mef*(A)	0 (0)
4	*erm*(B), *erm*(Q)	2 (2.67)
Tetracycline		
5	*tet A*(P)	13 (17.33)
6	*tet B*(P)	2 (2.67)
7	*tet A*(P), *tet B*(P)	16 (21.33)
8	*tet* (M)	0 (0)
9	*tet* (L)	0 (0)
10	*tet*(K)	0 (0)

Bacitracin		
11	*bcrA*	0 (0)
12	*bcrB*	3 (4.00)
13	*bcrD*	1 (1.33)
14	*bcrR*	1 (1.33)
15	*bcrA, bcrB*	0 (0)
16	*bcrB, bcrD*	1 (1.33)
17	*bcrD, bcrR*	0 (0)
18	*bcrA, bcrB, bcrD*	2 (2.67)
19	*bcrB, bcrD, bcrR*	1 (1.33)
20	*bcrA, bcrB, bcrR*	1 (1.33)
21	*bcrA, bcrB, bcrD, bcrR*	3 (4.00)
Gentamicin		
22	*aac(3)-II*	0 (0)
23	*aac(6')-Ib*	0 (0)
24	*aac (6')-II*	0 (0)
25	*ant(3'')-I*	1 (1.33)
26	*aph (3')-VI*	0 (0)
27	*arm A*	0 (0)
28	*rmt B*	0 (0)
Lincomycin		
29	*lnu*(A)	0 (0)
30	*lnu*(B)	0 (0)

Chicken and dogs shared the maximum number of isolates in this study and it was found that chicken isolates (n=18) had higher antimicrobial resistance genes compared to dog isolates (n= 11). Species wise presence of AMR genes are presented in Table 6.

Table 6. Species wise presence of AMR genes

Species	Gentamicin	Tetracycline	Erythromycin	Bacitracin
Chicken	0	13	4	4
Ostrich	0	0	1	0
Turkey	1	4	2	1
Guinea Fowl	0	2	0	2
Japanese Quail	0	2	1	2
Dog	0	5	8	0
Goat	0	3	3	1
Sheep	0	0	5	1
Pig	0	1	0	1
Cattle	0	0	0	1
Buffalo	0	1	2	0
Total	**1**	**31**	**26**	**13**

3.4. Association between phenotypic and genotypic antimicrobial resistance

Hierarchical clustering analysis using heat map analysis and correlation matrix was used to find the association between phenotypic and genotypic antimicrobial resistance among the different isolates recovered. Heat map analysis of *C. perfringens* isolates based on genotypic resistance profile showed the grouping of isolates from different animals into four main clusters. *C. perfringens* type A isolated from Guinea fowl and Japanese quail clustered separately (cluster I) based on the presence of bacitracin and tetracycline genes. Cluster II includes isolates from sheep (type C and D), buffalo (type F), chicken (type A), and turkey (type A). Type C and D isolates from sheep formed a sub-cluster in cluster II (Figure 3). Cluster III contains isolates from cattle (type A), dog (type A, C and F), pig (type A), and goat (type A). Cluster III has been grouped together based mainly on the resistance of the tetracycline gene followed by the bacitracin gene. It is interesting to note that all the isolates from dogs irrespective of the toxinotype were grouped in cluster III. Cluster IV include isolates from sheep (type D), buffalo (type A), goat (type D), ostrich (type A). Isolates in IV are clustered due to the presence of the erythromycin gene.

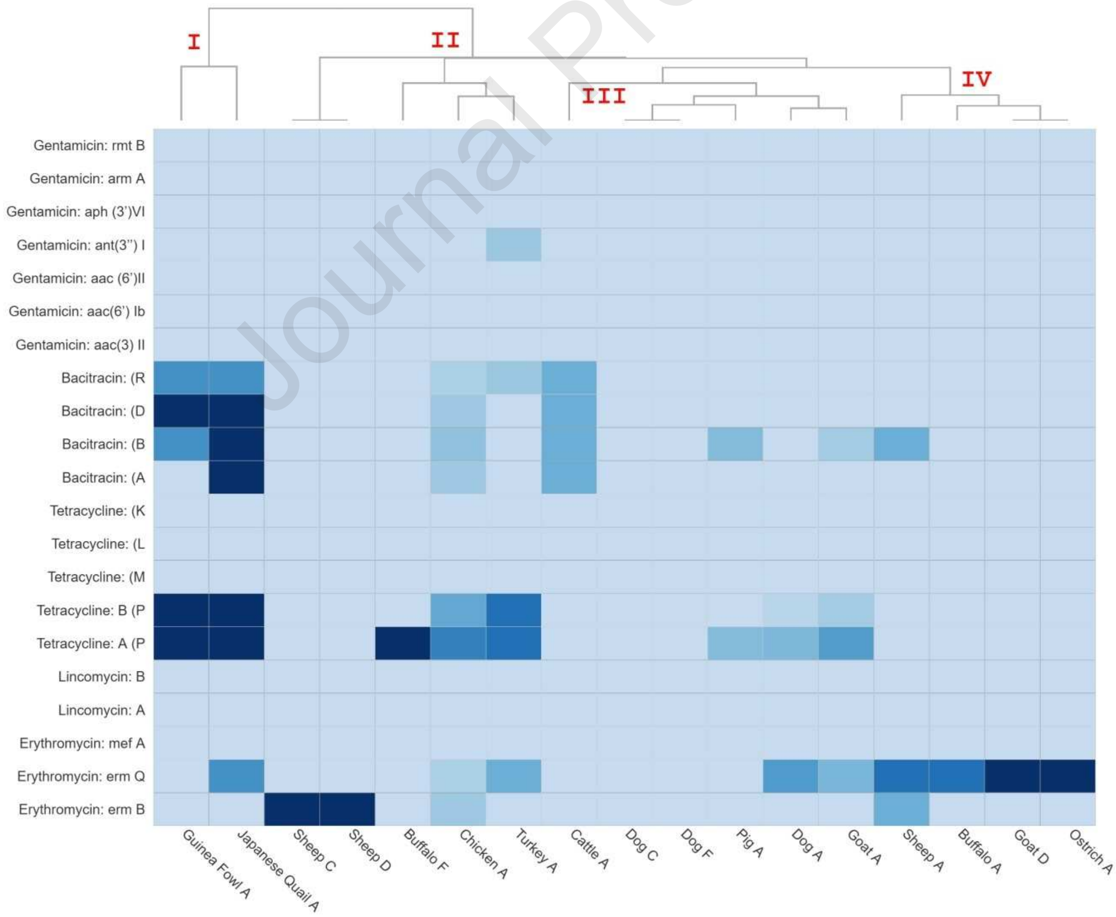

Fig.3. Hierarchical cluster analysis based on antimicrobial resistant genes. Lincomycin: A= *lnu*(A) gene; Lincomycin: B= *lnu*(B) gene; Tetracycline: A (P = *tet A*(P) gene; Tetracycline: B (P = *tetB*(P) gene; Tetracycline: (M = *tet*(M) gene; Tetracycline: (L = *tet*(L) gene; Tetracycline: (K = *tet*(K) gene; Bacitracin: (A = *bcrA* gene; Bacitracin: (B = *bcrB* gene; Bacitracin: (D = *bcrD* gene; Bacitracin: (R = *bcrR* gene. Four clusters (I, II, III, IV) are marked in red font.

Heat map analysis based on toxinotypes, phenotypic and genotypic profile of antimicrobial resistance revealed grouping of isolates into 4 clusters. Cluster I mainly comprises *C. perfringens* type A, one each from type C and F, grouped based on gentamicin and bacitracin resistance phenotypically (Figure 4). Cluster II encompasses *C. perfringens* type A with tetracycline resistance genes. Cluster III also comprises type A with tetracycline and bacitracin resistance genes. Cluster IV mainly contains type A along with 3 type D and 1 type A isolates grouped together based on erythromycin resistance (phenotypic and genotypic), tetracycline, gentamicin and bacitracin resistance phenotypically.

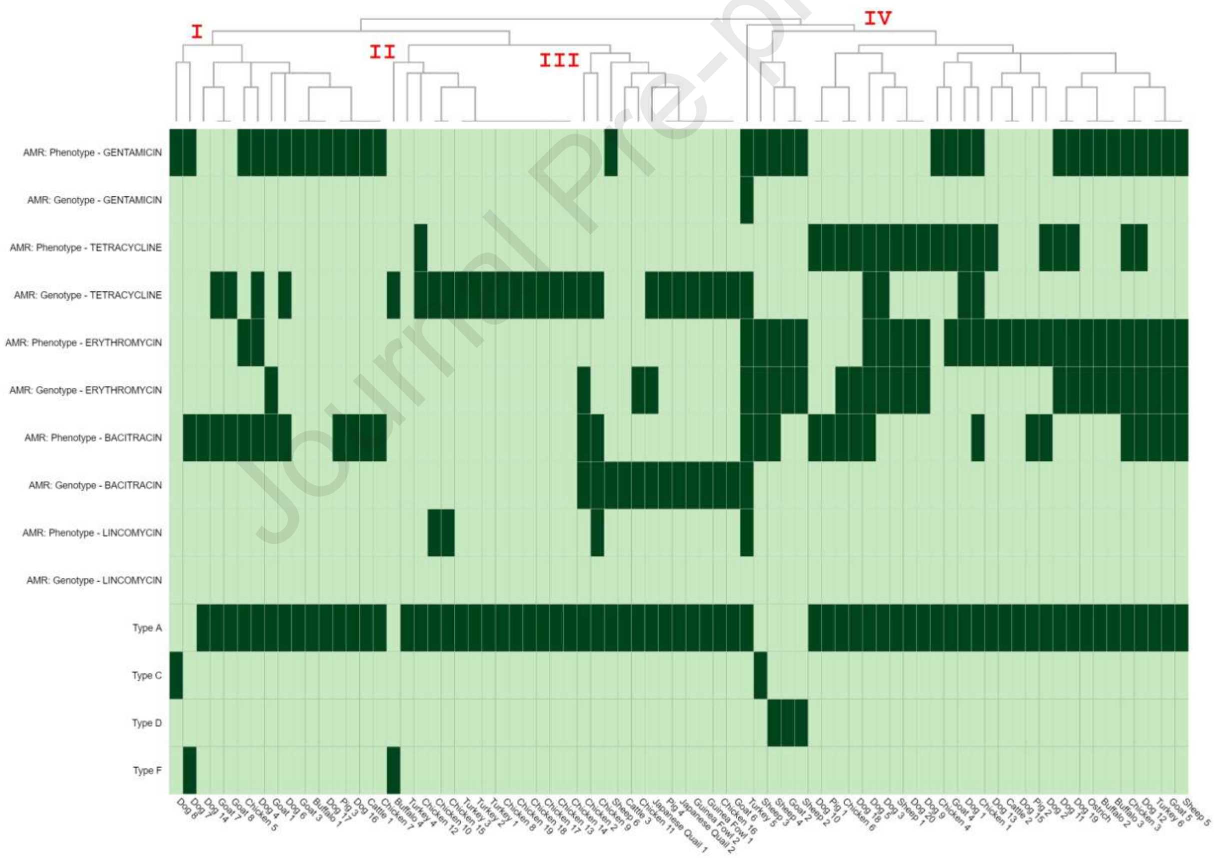

Fig.4. Heat map analysis based on phenotypic, genotypic resistance and type of *C. perfringens*. Four clusters (I, II, III, IV) are marked in red font

The correlation matrix analysis of genotypic resistance among isolates showed that there was a weak positive relationship between the tetracycline gene and bacitracin gene. This implies the co-presence of tetracycline and the bacitracin genes in many isolates. Contrarily, there was a weak negative relationship between the tetracycline resistance gene and erythromycin resistance gene (Figure 5) implying lesser co-presence of tetracycline and erythromycin genes among the isolates. When phenotypic and genotypic profiles of antimicrobials tested were analyzed there was a positive correlation for erythromycin while there was no correlation for other antimicrobials tested.

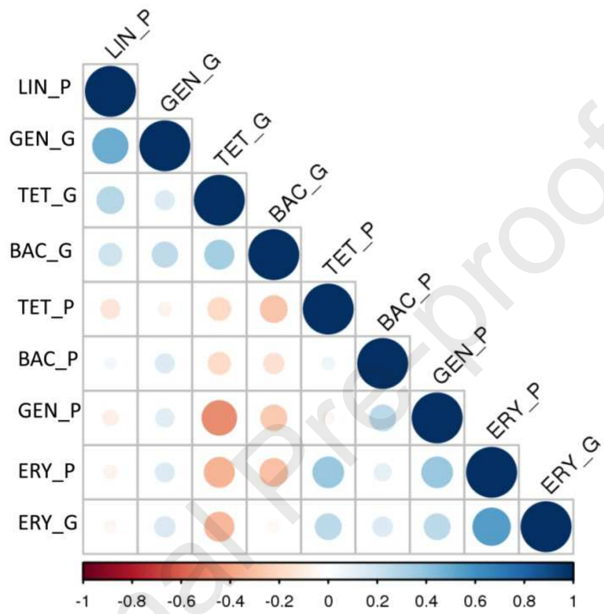

Fig. 5. Correlation matrix between phenotypic and genotypic antimicrobial resistance

4. Discussion

In the present study, an overall prevalence of 10.76% has been reported which is lesser compared to other earlier multi-species studies from India [12]. Samples were collected randomly and fecal samples from healthy animals constituted a major part of this survey. Multi-species (pig, goat, fish and human) studies from India involving mostly samples from diarrheic cases reported a higher prevalence of 36.09% [12]. Most of the *C. perfringens* recovered in the present investigation were type A (n=68) which was in accordance with earlier reports on the global dominance of type A [16-18]. Apart from type G, type A has also been implicated in necrotic enteritis in poultry and found ubiquitously in the intestine of animals [19, 20]. One isolate from a chicken showed the presence of *cpb2* gene which is linked with its role in gastroenteritis [21]. Type C has been reported to cause enterocolitis in several animal species [5] and the present study also showed the presence of type C in a dog and sheep sample. Type D, the causative agent of enterotoxaemia in small ruminants was recovered from 2 sheep and a goat, similar to the earlier reports [22, 23]. Type F (*cpa* and *cpe* gene) has been recovered from a dog and buffalo sample. *C. perfringens* with *cpe* gene was reported to be involved in food-borne poisoning and this enterotoxigenic organism was reported to range from 28.6 to 32.4% in India [12]. There is no molecular relationship of *cpe* gene with enteritis in animals but evidence shows that the presence of *cpe* gene is higher in diarrheic dogs than healthy dogs. *cpe* positive *C. perfringens* has been recovered from canine parvoviral enteritis in dogs [24]. c*pe* toxin may be involved in gastroenteritis in animals like foal necrotizing enteritis and canine hemorrhagic gastroenteritis still its role is questionable. *cpe* encoded in the chromosome has been linked to cause 70% of food-borne gastroenteritis in humans than the plasmid-encoded *cpe* which is responsible for non-food-borne gastroenteritis [25]. Thus, the type F isolates from the present study need further investigation to know its role in food poisoning in humans.

Phenotypically, gentamicin resistance was found to be 44% among the recovered isolates followed by tetracycline (40%) and bacitracin (40%). Similar high resistance to gentamicin, tetracycline and bacitracin was found by Mwangi et al. [7], Yadav et al. [12] and Udhayavel et al. [26] respectively. Most of the poultry isolates in the study are resistant to multiple drugs which may be due to improper use of antimicrobial drugs like bacitracin, tetracyclines, penicillin, gentamicin and streptomycin that are utilized for improving poultry production [6]. Strict anaerobic organisms are reported to be resistant to aminoglycoside like gentamicin [27]. Lincomycin resistance was found only in chicken and turkey isolates which may be correlated with the use of lincomycin for treatment of necrotic enteritis in poultry [7]. Although several reports have been documented on antimicrobial resistance among pathogens affecting livestock in India, genotypic characterization studies on antimicrobial resistance among *C. perfringens* in India are scarce.

Antimicrobial resistance gene profiling revealed a higher presence of the tetracycline resistance gene (41.33%) followed by erythromycin (34.66%) and bacitracin (17.33%). The tetracycline resistance gene is located on plasmid pCW3 containing *tetA*(P) gene coding for efflux protein responsible for resistance. *tetB*(P) is also located on the same plasmid, coding for a ribosomal protection protein, extended lower tetracycline resistance [28]. As reported earlier, the present study documented higher number of isolates with *tetA*(*P*) gene (n= 29) compared to *tetB*(P) gene (n= 18). Resistance due to tetracycline may be attributed to more than one resistance gene and the present study documented 16 isolates to possess both *tetA*(P) and *tetB*(P) genes [16]. None of the isolates in this study had *tet*(M), *tet*(L) and *tet*(K) genes. Similar to the present study, an investigation in canine species showed 96% of the tetracycline resistance was linked to *tetA*(P) gene while 41% of the isolates had both *tetA*(P) and *tetB*(P) genes and there was no presence of *tet*(M) gene [29].

erm(Q) gene was found in 21 isolates and *erm*(B) in 7 isolates. Soge et al. [30] showed that *erm*(B) is the common macrolide resistance gene in *C. perfringens* while Berryman et al. [31] reported *erm*(Q) as the resistance gene. *mef*(A) was not found in any isolate. No isolates had lincomycin resistance genes (*lnu*A and *lnu*B). Reports documented that lincomycin resistance is conferred by macrolide-lincosamide-streptogramin B resistance-conferring gene *erm*(B) and *erm*(Q). Resistance to lincosamide and macrolide antimicrobial is well documented in *C. perfringens* hence, these drugs are not the first choice drugs for *C. perfringens* [32].

Bacitracin resistance is commonly found in avian strains of *C. perfringens* since this antimicrobial is used to improve production [33]. In the present study, 9 poultry isolates had bacitracin resistance genes (individual or in combination) while 4 isolates were from other animals. Bacitracin resistance is conferred by *bcrRABD* locus that is responsible for the efflux of bacitracin from the cells [33]. *bcrA* and *bcrB* genes are essential to confer bacitracin resistance as indicated through mutation studies of both these genes. *bcrD* and *bcrR* resistance genes were identified in 8 and 6 isolates respectively but, there is no clear role of these genes in the resistance of bacitracin as reported by Han et al. [33]. The gentamicin resistance gene was present in one turkey isolate. To the best of our knowledge, there is no published report on the use of PCR for gentamicin resistance determination in *C. perfringens*. Hence, as an initial step, primers used for the detection of gentamicin in the genus Enterobacteriaceae were used in the present study. Moreover, it is reported that anaerobic organisms might have reduced intracellular transport of aminoglycoside antimicrobials hence, these antimicrobials may not act on *C. perfringens*. *C. perfringens* might have evolved a mechanism to resist aminoglycoside antimicrobials [27]. Reports also documented the susceptibility of *C. perfringens* to gentamicin at higher concentrations [26]. Hence, the genetic characterization of aminoglycoside resistance genes in *C. perfringens* needs to be elucidated by whole genome sequencing.

Several isolates though showed the presence of antimicrobial resistance genotypically but were found to be sensitive phenotypically which may be due to lack of expression of the resistance gene or point mutation of these genes [16]. There were certain situations where phenotypically the isolates were resistant to some antibiotics but the available resistance genes could not be amplified. Additional genes may be involved in conferring resistance against these antimicrobials hence, needs whole-genome analysis of the isolates [34]. Studies targeting both phenotypic and genotypic relatedness among *C. perfringens* needs to be carried out to map the association between genes involved and antimicrobial resistance exhibited phenotypically.

The correlation matrix showed a weak positive correlation between tetracycline and bacitracin resistance genes. The tetracycline gene is located on a conjugation plasmid like pCW3 and bacitracin resistance is located on conjugation plasmid and transposons. The magnitude of bacitracin resistance gene transfer was less when conjugation plasmid conferring bacitracin resistance was absent and only mobile elements were present. It has been reported that conjugation plasmid is responsible for the maintenance and transfer of mobile elements including bacitracin elements [28]. Hence, plasmid profiling and conjugation experiments can provide insights into the role of a possible mechanism of plasmid-mediated transfer of resistance gene. Similarly, there was a weak correlation between phenotypic and genotypic resistance among different isolates.

Due to the ease of the disc diffusion method, it is employed for AST of various organisms but they are not accurate for anaerobic organisms [15]. The use of the disc diffusion method for AST is one of the limitations of this study. To minimize the error caused by the phenotypic method, the molecular method was employed to identify the resistance genes. Details regarding the use of antimicrobials in the animals sampled and health status during the time of sampling are not complete. These details might have enlightened the reason for the presence of AMR genes among the isolates recovered. Further to know the role of different genes in antimicrobial resistance and the mechanism involved in the transfer of resistance genes, whole-genome sequencing of the resistant isolates can be performed.

5. Conclusions

Antimicrobial resistance is a burning problem worldwide and resistance is recorded among various anaerobic organisms including *C. perfringens*. The present study documented the phenotypic and genotypic antimicrobial resistance among *C. perfringens* isolates of multi-species origin. *C. perfringens* type A, C, D and F were isolated during the study and multidrug resistance was recorded among several isolates. Multidrug-resistant genes were also present in the study showing the impact of antimicrobial resistance among pathogens like *C. perfringens* that are ubiquitously present. Studies documenting both phenotypic and genotypic antimicrobial resistance are needed further to improve the available data on the genes involved in resistance. The study warrants immediate surveillance schemes to control the irrational use of antimicrobial agents in animal production to break the spread of antimicrobial resistance.

Declaration of competing interest

Authors declare no conflict of interest.

Acknowledgements

Authors like to acknowledge the funding received from DBT for the Twinning project on **"**Genetic "Characterization of Antibiotic Resistance in *Clostridium perfringens* and *Clostridium difficile* of food animal origin and its Public Health Significance**"** **(**BT/PR25444/NER/95/1200/2017, 23.05.2018).

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Highlights

- *C. perfringens* type A (90.67%), type C (2.67%), type D (4%), type F (2.67%) were recovered.
- Genotypic characterization of the antimicrobial resistance showed that tetracycline (41.33 %), erythromycin (34.66%) and bacitracin (17.33%) resistance genes
- Correlation matrix showed weak positive relationship between tetracycline gene and bacitracin gene and weak negative relationship between tetracycline resistance gene and erythromycin resistance gene

Conflict of Interest Statement

- None of the authors of this paper has a financial or personal relationship with other people or organizations that could inappropriately influence or bias the content of the paper.
- It is to specifically state that “No Competing interests are at stake and there is No Conflict of Interest” with other people or organizations that could inappropriately influence or bias the content of the paper.